

ELECENG 3CL4: Introduction to Control Systems

Lab 4: Phase Lead Compensation Using Root Locus

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Objective

To use the root locus technique to design a phase lead compensator for a marginally-stable servomotor.

Assessment

This laboratory is conducted in groups of no more than two students. Students are required to attend their assigned lab section. The assessment of this lab will occur based on your answer to the pre-lab questions, in-lab activities, and a written laboratory report. **Each group is required to submit their answers to the pre-lab questions as a hard copy at the beginning of their designated lab session to the Teaching Assistants (TAs) on duty.** No marks will be awarded to pre-labs submitted 10 minutes after the designated start time of the lab session. Only one submission is necessary per group. **Both group members should clearly state their full names, student numbers, Lab section, and Class section(s) on the title page of the pre-lab answers document.** It is up to you if you want to submit a handwritten document or a printed one. You will earn a maximum of 100 marks from Lab 4 activities. **Lab 4 will constitute 5% of your total grade for this course.** The components of the assessment are as follows:

- Pre-lab: Root locus based phase lead control design and evaluation (Section 1; 35 marks);
- In-lab design and experiment activities (Sections 2 and 3; 40 marks);
- Final laboratory report (Section 4; 25 marks).

Your performance in the experiments will be evaluated during the lab by the TAs. The marks for each component are clearly indicated in this lab manual.

1 Pre-lab Design Exercise

The purpose of this pre-lab design exercise is to prepare you for the actual phase lead controller design during the lab. The plant (process) model parameters and design requirements in the pre-lab are somewhat different from those in the actual lab, but the design steps are essentially the same.

Consider the feedback control system in Figure 1 where $G(s) = \frac{4.7}{s(s+3.2)}$, $H(s) = 1$, $N(s) = 0$, and $T_d(s) = 0$.

The goal is to design a phase lead compensator of the form $G_c(s) = k_c \frac{s+z}{s+p}$, $p > z$, so that for a unit step input the percentage overshoot of the output $y(t)$ is 10% and its 2% settling time is 1 second. Follow the the control system design steps below to achieve these objectives:

1. **(2 marks)** Compute the desired closed-loop pole locations based on the overshoot and settling time requirements given above.

We would like to acknowledge the efforts of Dr. Ayman Negm in the development of this lab.

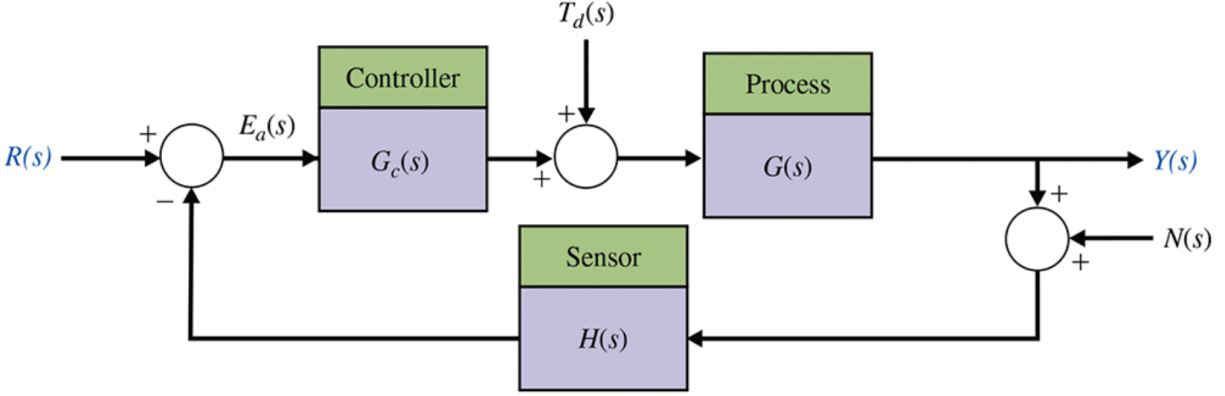


Figure 1: Block diagram of a generic feedback control system. We will focus on the case in which $H(s) = 1$, noise is negligible ($N(s) = 0$) and there is no disturbance ($T_d(s) = 0$) (Figure 4.4 of Dorf and Bishop, *Modern Control Systems*, 14th edition, Pearson, 2022). Note: This figure is in the Laplace domain and $Y(s) = \Theta(s)$.

2. **(6 marks)** Assuming a proportional controller, i.e., $G_c(s) = k_c$, sketch the root locus of the closed-loop control system for $0 < k_c < +\infty$. Is there a value of k_c that would place the closed-loop poles at the desired locations?
3. **(4 marks)** Let s_0 be one of the desired dominant closed-loop poles. Compute the angles of $G(s_0)$ and from it the required angle contribution for the controller, $\phi_c = \angle G_c(s_0)$, so that s_0 can actually be a closed-loop pole for some value of k_c . Explain how a phase lead compensator can help provide this required phase.
4. **(6 marks)** Compute the values of z and p of the phase lead compensator based on ϕ_c in the previous step. Note that given the two free design parameters and only one condition to satisfy, you need to choose the value of one of these parameters. In the design of phase lead compensators, the designer usually selects the value of z and computes p accordingly. Let $s_0 = \sigma + j\omega$ denote a dominant closed-loop pole. A typical choice for z is σ . Now find the value of p such that

$$\angle G_c(s_0) = \angle(s_0 + z) - \angle(s_0 + p) = \phi_c.$$

5. **(7 marks)** Sketch the root locus of the closed-loop system as a function of the lead compensator gain k_c . Note that in this case the relevant part of the open-loop transfer function is $P(s) = \frac{s+z}{s+p} \cdot \frac{1}{s(s+3.2)}$.
6. **(4 marks)** Use the magnitude condition, i.e., $|G_c(s_0)G(s_0)| = 1$, to compute the value of k_c that would place two of the closed-loop poles at the desired locations.
7. **(4 marks)** Compute the closed-loop transfer function $T(s) = \frac{G_c(s)G(s)}{1+G_c(s)G(s)}$ and comment on its form. Do you expect any deviation from the design objectives? Explain.
8. **(2 marks)** Compute the velocity error constant for the feedback control system that you have designed as well as the steady-state error to a unit ramp input. Recall that for a stable feedback control system, the velocity error constant is defined as $k_v = \lim_{s \rightarrow 0} sG_c(s)G(s)$. The steady-state error to a unit ramp input is given by $e_{ss} = \frac{1}{k_v}$.

Your pre-lab report must include all relevant computations and sketches.

2 Phase-Lead Control of the Servomechanism (20 marks)

In Lab 3, we observed that the gain of a proportional controller could be chosen to achieve a desired overshoot in the step response of the closed-loop servomechanism control system. However, we were unable to adjust the settling time of the closed-loop by manipulating the value of the gain. Later, we designed a proportional controller with velocity feedback to eliminate this limitation of the proportional controller.

In this lab, we will design a *phase lead compensator* that will enable us to place the dominant poles of the closed-loop system at some desired locations in the s -plane. In particular, we will choose the locations of the poles to achieve a desired overshoot and settling time for the step response of the closed-loop control system. We will also examine the velocity error constant obtained in this design.

2.1 Design of Phase-Lead Compensator

1. **(10 marks)** Follow the procedure in the pre-lab exercise to **design a phase lead compensator for the DC motor servomechanism of the lab to achieve a 10% percentage overshoot and a 2% settling time of 0.3 sec in the step response of the closed-loop control system.** Recall that you have already identified a model of the plant $G(s)$ in your Lab 2 experiment. If you are using a different equipment setup this time, you can ask your TA to provide you with the model of the system for the purpose of this lab. [Optional: Use the MATLAB *rlocus* command to verify your work.]
2. **(2 marks)** Compute the velocity error constant, k_v with your designed controller.

(Hint: The following MATLAB commands may be used to complete steps 3 and 4: **step**, **feedback**, **tf**, **zero**, **pole**. In the MATLAB command line, type *doc step* to open MATLAB documentation on how to use this command. You can similarly refer to MATLAB help to see how all of these commands are to be used.)

3. **(5 marks)** With the help of MATLAB, compute the closed-loop transfer function $T(s) = \frac{G_c(s)G(s)}{1+G_c(s)G(s)}$ and its poles and zeros.
4. **(3 marks)** Using MATLAB, plot the step response of the closed-loop system. Does your control system satisfy the design objectives? Comment on any potential discrepancy in the results.

At this point, you should call the Teaching Assistant to have your Section 2 work evaluated. **Note:** To obtain marks for this section, you must show a TA your worked solution, including the computed position for the desired closed-loop poles, the final form of the lead controller designed for your process (i.e., the motor at your workstation with its parameters A and τ_m that you determined in Lab 2), and the computed value of k_v . You also need to demonstrate, using MATLAB, the closed-loop transfer function and its poles and zeros, as well as the closed-loop step response and discuss whether the design requirements have been satisfied. **Provide the work done in this section, in response to each of the four points above, in your final lab report and include any analysis and plots, if required.**

3 Experiment with Phase Lead Compensator (20 marks)

The experiments in this section will examine the step and ramp responses of the closed-loop system with the designed controller in the previous section. You should follow the steps below to carry out the experiments.

3.1 Step Response (12 marks)

- i) Download the *Simulink starter file* from Avenue (Content > Laboratory > Simulink Starter File > EE3CL4_Lab.slx). You should create a separate folder for this lab on the computer. Place the starter file in the Lab 4 folder and name it appropriately (e.g., Lab4_Exp1).
- ii) Open MATLAB, then Simulink, and open your Lab 4 Simulink file that was saved above.
- iii) Find the *HIL Write Analog*, *HIL Read Encoder*, *Pulse Generator*, *Sum*, *Gain*, *Scope*, *To Workspace* and *Transfer Fcn* blocks from the Simulink library and add them to the model.
- iv) Follow Figure 1 to build the feedback control system in Simulink environment. As mentioned in the figure caption, $H(s) = 1$ (replaced by a wire) and $N(s) = 0$ (noise is negligible). For now, we will also not consider the disturbance $T_d(s)$, i.e., $T_d(s) = 0$.
- v) Model the *reference input* ($r(t)$) in time domain, which is shown as $R(s)$ in Laplace domain in Figure 1) by using a *Pulse Generator* block and set its *Amplitude*, *Period* and *Pulse Width* to 1.5, 6 and 50, respectively. This would set up the reference input $r(t)$ as a pulse train with amplitude of 1.5 radians or 85.94° , period of 6 seconds, and duty cycle of 50%.
- vi) Open the *Transfer Fcn* block and enter the parameters of the transfer function of the phase lead compensator that you have designed in Section 2. For example, if the transfer function is of the form $G_c(s) = \frac{k_c(s+a)}{(s+b)} = \frac{k_c s^1 + k_c a s^0}{s^1 + b s^0}$, then you will enter the *Numerator coefficients* as $[k_c \ k_c a]$ and the *Denominator coefficients* as $[1 \ b]$.
- vii) Open the *Gain* block at the output of the *Read Encoder* block which is meant to convert *Counts to Radians* (see Figure 2). Set its gain so that it receives *Counts* at its input and outputs position in *Radians* (Hint: You have already done this in Labs 1, 2, 3).
- viii) We will run the motor only for 6 second intervals in this experiment. To do this, go to the *Modeling* tab in the Simulink window and then click on *Model Settings*. In the *Solver* tab and in the *Simulation time* area, set the *Stop time* to be 6 seconds. Click *Apply* and then OK. Save your Simulink model.
- ix) When you are done building your model by following Figure 1, it should look like the target model shown in Figure 2. **STOP and VERIFY the following (Failure to do so may lead to damaging your motor):**
 - Have you converted *counts to radians* at the output of the *Read Encoder* block? (Step vii above)
 - Make sure that the closed-loop is set as a negative feedback loop.
- x) Turn ON the Qube-Servo 3 motor block using its ON/OFF button. Align the engraved line on the red circular load with the 0° mark at the top of the Qube-Servo 3.
- xi) In the Simulink window, go to the *Hardware* tab and press *Build* (found in the *Build, Deploy & Start* drop-down menu) to generate the Simulink model and then press the *Monitor and Tune* button to initiate the experiment.

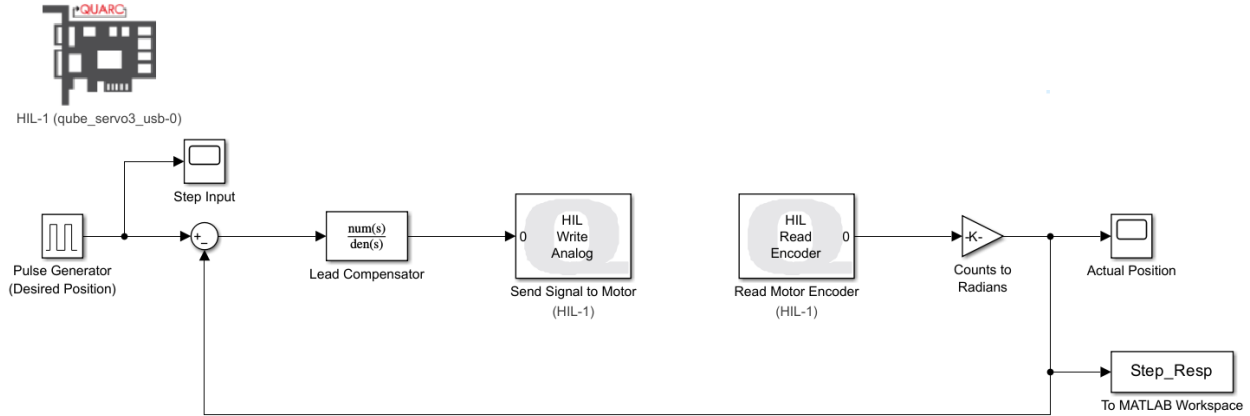


Figure 2: Target Simulink Model for the Step Response Experiment with Lead Compensator.

- xii) Send your data to the MATLAB workspace by using the *To Workspace* block and save it for at least one complete step response period so that you can use it in your lab report. You will need to use the *save* command in MATLAB to do so. Ask the TAs for help, if required.
- xiii) Verify that the control system design objectives as stated in Section 2.1 Point 1 (percentage overshoot of 10% and 2% settling time of 0.3 sec) are satisfied for the step response (use the cursors in the Simulink scope or work in MATLAB workspace). **Discuss this in your lab report with associated plot(s).**
- xiv) Now replace the *Transfer Fcn* block with a *Gain* block and enter the value of k_c , i.e., replace your lead compensator with a proportional controller with the same gain as the lead compensator k_c . (Note: You may want to make a copy of your lead compensated model's Simulink file first.)
- xv) Press *Build* to generate the Simulink model and then press the *Monitor and Tune* button to initiate the experiment.
- xvi) Send your data to the MATLAB workspace by using *To Workspace* block and save it for use in your lab report.
- xvii) Compare the step response with and without the lead compensator pole and zero. Are any of the design objectives met with only the proportional controller? **Include your analysis and comments in the lab report and accompany them with associated plot(s).**
- xviii) Turn OFF the Qube-Servo 3 motor.

At this point, you should call the Teaching Assistant to have your Section 3.1 Step Response work evaluated. **Note:** To obtain marks for this section, you must show a TA that your lead compensated model fulfills the percentage overshoot and 2% settling time design requirements. You also need to demonstrate the step response with only the proportional controller to your TA and discuss whether it fulfills any of the design requirements. **Your findings must also be included in the final report. See Points (xiii) and (xvii) above.**

3.2 Ramp Response (8 marks)

- i) Copy the Simulink model of your **lead compensated system** from Section 3.1 under a different name. Ensure that this is the model with the lead compensator that you designed in Section 2 (not the proportional controller). You will use this model as a base for the ramp response experiment.

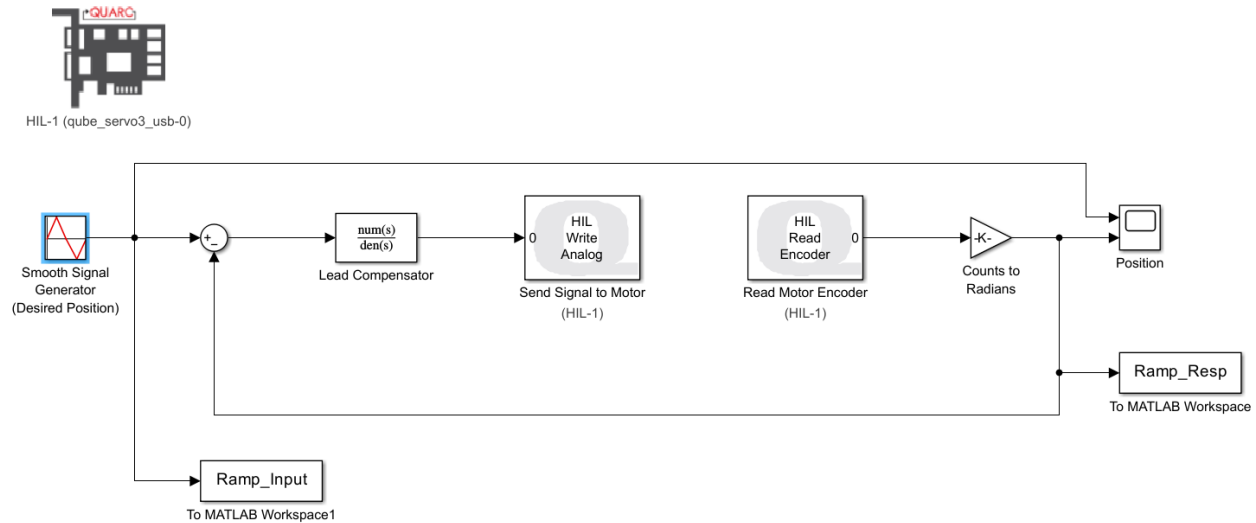


Figure 3: Target Simulink Model for the Ramp Response Experiment with Lead Compensator.

- ii) Open the Simulink model that you saved above. Find the *Smooth Signal Generator* block (you can search for it in the Simulink library) and use it to replace the *Pulse Generator* block in the model. You can delete the *Pulse Generator* block.
- iii) Open the *Smooth Signal Generator* block. Select the “Triangle” wave as the signal type. Set the *Amplitude*, *Frequency* and *Initial phase* to 1.5 radians, 1 Hz and $\pi/2$ radians, respectively.
- iv) When you are done building your model it should look like the target model shown in Figure 3.
- v) Verify that the *Stop time* is set to 6 seconds in the *Model Settings*, *counts* are being converted to *radians* at the output of the *Read Encoder* block, and the closed-loop is set as a negative feedback loop.
- vi) Turn ON the Qube-Servo 3 motor block using its ON/OFF button. Align the engraved line on the red circular load with the 0° mark at the top of the Qube-Servo 3.
- vii) Press *Build* to generate the Simulink model and then press the *Monitor and Tune* button to initiate the experiment.
- viii) Send your data to the MATLAB workspace by using the *To Workspace* blocks (both the ramp input and the ramp response) and save it for at least one complete ramp response period so that you can use it in your lab report.
- ix) In Section 2.1 (Point 2), you theoretically computed the velocity error constant, k_v . Use that value of k_v to find the value of steady-state error (e_{SS}) that we expect. Now, measure the values of the velocity error constant k_v and the steady-state error e_{SS} . Do the measured values match the theoretical expectations? **Discuss this in your lab report with associated plot(s).** While doing this part, the following procedure maybe helpful:
 - Plot both the ramp input and output (ramp response) on the same scope (see Figure 3). This will make it easy for you to calculate the steady-state error using cursors on the scope.
 - The steady-state error (e_{SS}) between the ramp input and output (ramp response) is computed once the response has sufficiently settled and is the vertical distance between the two lines, i.e., between the ramp input line and the ramp response line.

- You can measure the velocity error constant k_v by first measuring the steady-state error e_{SS} (as noted above) and then using the formula $e_{SS} = \frac{A}{k_v}$. You can find the slope of the ramp input, A , by using cursors to pick two points on the ramp input, i.e., (x_1, y_1) and (x_2, y_2) , and then use the *point-slope formula*. With e_{SS} measured and slope A determined, you can find the *measured* value of k_v as $k_v = \frac{A}{e_{SS}}$.

x) Turn OFF the Qube-Servo 3 motor.

At this point, you should call the Teaching Assistant to have your Section 3.2 Ramp Response work evaluated. **Note:** To obtain marks for this section, you must show a TA your ramp response and your measured k_v and e_{SS} values. **Your findings must also be included in the final report. See Point (ix) above.**

4 Laboratory Report (25 marks)

Each group (of two students) must submit an electronic lab report through the Avenue to Learn Dropbox. Only one student in each group needs to submit the report to the Avenue to Learn Dropbox. The other student should verify the submission. **The report is due by 11:59 pm the day after the students' designated lab session.** For example, if your lab is on February 4, 2026, then your lab report would be due on February 5, 2026 at 11:59 pm. The report should be submitted as a PDF and formatted in single-column, single-spaced, using Times New Roman 12 or equivalent font. The group members should clearly state their individual contributions to the report in a statement in the beginning of the report. **Both group members should clearly state their full name, student numbers, Lab section, and Class section(s) on the title page of the report.** The report file should be named as: 3CL4_Lab_A_BB_Student1_Student2.pdf, where A is the lab number, BB is the lab section that the students belong to, *Student1* is the nine digit McMaster student number of the first student and *Student2* is the student number of the second student. **A 10% penalty will be applied if the above-mentioned title page and/or file naming conventions are not followed.**

The laboratory report must include the design calculations and plots of the results of the experiments, followed by a brief analysis of these results. You must refer to the statements mentioned in bold and in dark blue color in Sections 2 and 3 to determine what to include in the lab report. You must compute the values of settling time and percentage overshoot for the step response, and steady-state error for the ramp response from the results of the experiments and compare those to the corresponding theoretical values. You should discuss any potential discrepancies between the experimental and theoretical results.